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Live vs. Static Comments: Empirical Analysis of Their Differential Effects on User Evaluation of Online Videos

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Abstract: Unlike static comments, which are typically located below online videos, live comments affect consumers' video-watching experiences in real time and may influence their evaluation of the video in distinct ways. Despite the significance of live comments, few studies have explored the differentiated effects of live comments vs. static comments on user evaluation of online videos. Utilizing a dataset comprising approximately two million pieces of textual data from a leading Chinese online video platform, our findings reveal substantial differences between the effects of live and static comments, with these effects varying by video type (informational versus emotional) and showing notable changes during health threats. This study examines the differential impact of live vs. static comments, providing empirical evidence for distinct information processing pathways under varying time constraints. Our results shed light on the underlying mechanisms responsible for the diverse effects of different forms of social interaction, offering valuable theoretical insights. They also have managerial implications regarding how online video platforms can facilitate engagement among viewers and between video creators and their audiences.

Keywords: live comments; static comments; dual system theory; informational videos; emotional videos; health threats

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1. Introduction

Live comments, a novel form of social interaction distinguished by real-time interaction and social engagement, have experienced a dramatic surge in popularity in recent years, attracting a large and active user base that engages and contributes [1]. As of June 2024, the total number of live comments on the Chinese online video platform Bilibili had surpassed 20 billion [2]. As an integral part of the video-watching experience, live comments give users an avenue for quasi-social interaction and an additional channel for information acquisition, owing to their real-time interaction beyond the video content [3]. While watching videos, users can also gain information from live comments displayed on the screen, potentially triggering emotional resonance [4,5]. Live comments can also prompt immediate user reaction, encouraging the expression of thoughts and emotions triggered by each other's comments while watching online videos. This interactive dynamic not only provides additional information from other users but also evokes emotional responses, which can affect users' overall evaluation of the videos, distinguishing it from the more static form of traditional comments.

On the other hand, static comments, such as consumer reviews, are generally posted by users in a specific location [6,7]. As a typical form of static user-generated content, these comments also provide a platform where users can express their opinions and emotions

regarding the videos you have watched, offer detailed feedback on various elements of the videos, and engage in discussions with other users about the videos.

While much existing research has demonstrated the substantial impact of the latter form of social interaction (i.e., static comments) on users' attitudes toward videos [8–10], few studies have examined the former form of social interactions (i.e., live comments) [11–13]. Moreover, given the distinct characteristics of these two forms of social interaction, such as content, length, form, emotional expression, and dynamics, they may have significantly different influences on users' evaluations of the videos they have watched. However, no existing research has investigated their differential impacts.

This study aims to bridge this gap by addressing two key research questions: (1) whether and how do live comments and static comments differ in their impact on user evaluation of online videos; (2) whether and how these differential impacts vary under different conditions (e.g., video types and external environment). We propose that live comments, which appear dynamically on the screen during video playback, and static comments, typically displayed in a fixed area detached from the video content [14], impose different time constraints on users when processing these comments. This major distinction prompts users to adopt different information processing strategies. For instance, Pettibone [15] found that time pressure reduces the likelihood of individuals investing significant cognitive effort in making accurate product choices. Consequently, the distinct time constraints associated with live and static comments may lead users to employ different cognitive modes when processing information. Drawing upon dual system theory [16–18], we conjecture that users may process live comments through System 1, characterized by rapid, emotional, automatic, and intuitive responses. In contrast, when encountering static comments, users are more likely to engage System 2, enabling a thorough and rational evaluation of both positive and negative comments. These significant differences in information processing strategies can result in varying impacts of live and static comments—both in terms of positive and negative valence—on user evaluations of online videos.

Additionally, the impact of live and static comments may be moderated by internal and external factors, such as video types and the external environment. Generally, users watch online videos to achieve two primary goals: information-seeking and emotional satisfaction, respectively [19]. The former involves seeking or sharing information, while the latter pertains to emotional relaxation. As users engage with videos to fulfill these objectives, they may enter different cognitive states, which in turn influence how they process live and static comments. Accordingly, we categorize online videos into two types—informational and emotional—and aim to investigate the moderating effects of video types on the differential impact of live and static comments.

Our specific focus on the external moderator centers on health threats, which can significantly alter individuals' emotional states, thereby potentially influencing users' information processing strategies. Health threats can evoke negative emotions by diminishing individuals' perceived control over their health, which in turn affects their willingness to invest cognitive effort [20,21]. For example, the COVID-19 pandemic can provoke a range of emotional reactions, notably depression and anxiety [22]. When experiencing such negative emotions, individuals often seek greater emotional regulation, which can influence the cognitive effort they are willing to invest. Moreover, measures like social distancing and home isolation implemented during health threats have increasingly shifted social activities online [23]. Consequently, we anticipate that during periods of significant health threats, these threats will meaningfully moderate the differential impact of live and static comments.

To this end, we collected data from Bilibili, a popular Chinese social media platform known for its vibrant live comment feature and static comments, which provide an excellent

context for investigating the differential impact of live and static comments. Our dataset includes 667 videos, 1.59 million static comments, and 570,000 live comments, spanning September 2019 to July 2020. We employed a pre-trained ERNIE text analysis model to determine the sentiment polarity of both live comments and static comments. Our empirical analysis reveals several interesting findings. First, we show that both positive live and static comments positively influence users' evaluations of online videos, as measured by the number of virtual coins that users donate to video creators, and the positive impact of static comments is relatively stronger than that of live comments. Most intriguingly, we find that negative static comments also have a positive effect on user evaluation, whereas negative live comments have an insignificant effect. Second, we discovered that the positive effect of negative static comments is more pronounced for informational videos compared to emotional videos. Third, during health threats, our findings indicate that the positive impact of live comments becomes more significant than that under the environments without health threats. These findings underscore the distinct roles that live and static comments play in shaping online user behavior. They provide insights into the different information processing strategies that users employ when interacting with live versus static comments. Furthermore, the effects of these comment types vary across video categories and under different environmental conditions, specifically the presence or absence of external health threats.

Our findings contribute to the literature on social interaction and information processing. We are among the first to investigate the differential impact of live and static comments. Drawing on dual system theory, we explore how time constraints in live and static comments drive users to rely on different cognitive systems for processing, examining the boundary conditions of this effect by considering video type and the presence of health threats contexts. These findings not only provide empirical evidence for information processing under different time constraints but also offer valuable insights for marketers and video creators in quality control and evaluation of marketing effectiveness.

2. Theoretical Development

2.1. Live Comments and Static Comments

Comments on online videos serve as a medium for social interaction, providing users with a platform to express their perspectives on video content. Text comments are characterized by generous word limits and varied expression forms, enabling users to recall their experiences, convey emotions, offer suggestions, or seek social support [24]. In contrast to traditional static comments, live comments foster interactions at specific moments in a video. Originating in Japan, this dynamic form of engagement has rapidly gained popularity, especially in various Asian countries [25,26]. As shown in Figure 1, live comments are displayed above the video content and move from right to left on the screen in real time. These comments provide users with instantaneous information, entertainment, and social connections while they are immersed in video consumption [3,27]. This enhances the overall viewing experience, marking a shift in the landscape of online engagement associated with video content. Recently, an increasing number of US companies have adopted this technology to enrich users' video-watching experiences [28].

Figure 2 depicts a common layout of static comments, where the content is presented in a fixed region of the interface. As a prevalent form of user reviews, static comments have been extensively examined in the literature. Recently, there has been growing interest in investigating live comments, with a particular focus on their effects. Studies have shown that the informative content in live comments positively influences users' engagement, knowledge acquisition, and post-consumption appreciation of the product [4,12,13]. Psychological studies have shown that live comments fulfill users' emotional needs and

enhance their empathy [29]. Few studies, however, have considered the differences between live comments and static comments and separately explored their effects on user behavior. Moreover, there is a dearth of research on whether and how such effects differ under various circumstances (e.g., video type and health environment). To address these research gaps, we first elucidate the distinctions in the effects of live comments and static comments on users' evaluation of online videos. Then, we explore these distinctions across various video types and external factors (e.g., health threats).

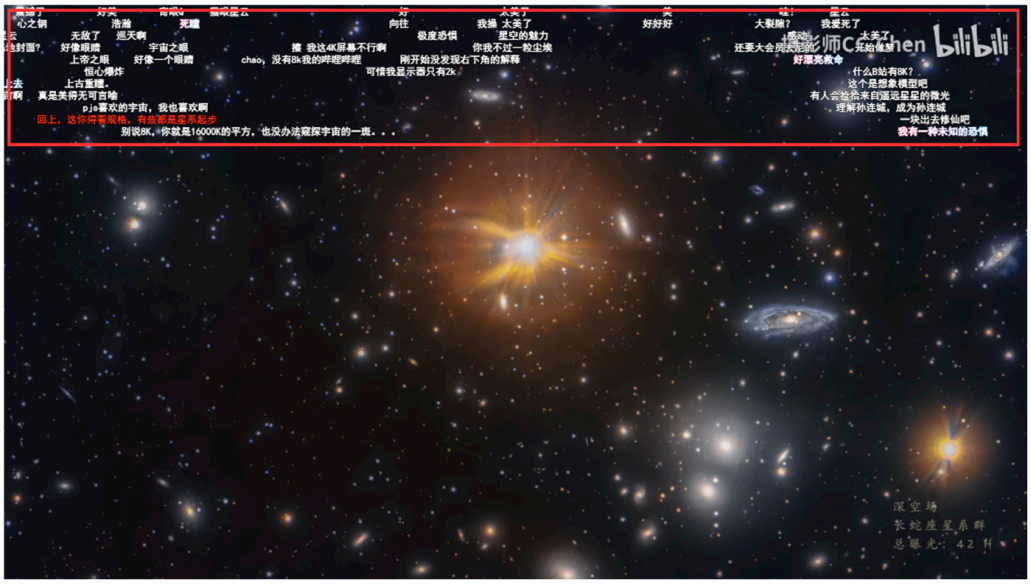

Figure 1. Visualization of live comments.

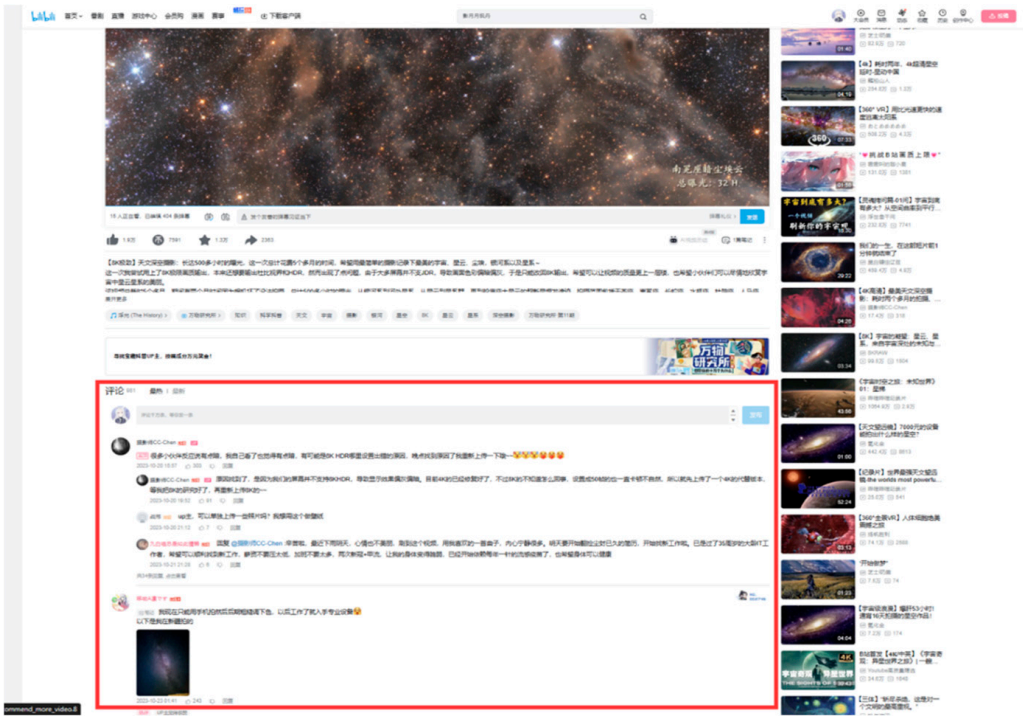

Figure 2. Visualization of static comments.

2.2. Dual System Theory

Based on dual system theory, psychologists have delineated two distinct modes of thinking, known as System 1 and System 2 [16–18]. System 1 operates swiftly, emotionally,

automatically, and intuitively, serving as the default mode for handling routine tasks and familiar activities [30]. This mode requires minimal conscious effort to process available information. If the focal problem is under specific time constraints, System 1 guides handling it. Individuals often rely on mental shortcuts or heuristics, such as previous life experience or patterns, to make rapid decisions [31]. However, System 1 thinking might sometimes be susceptible to errors or cognitive biases because it does not involve deep thinking or rigorous analysis [32,33].

System 2 is deliberate, conscious, analytical, and rational [34]. Under System 2 thinking, individuals typically take more time to process information and make decisions [35]. This mode supports abstract hypothetical thinking and active mental engagement, distinct from the rapid, intuitive nature of System 1 [36]. Different from System 1, System 2 thinking uses logic, systematic evaluation, and critical thinking to digest information [37]. When individuals are more dependent on System 2 thinking, they might make better, more accurate decisions.

Both System 1 and System 2 exist in individuals' minds when they process information and make judgments [38]. When individuals rely heavily on System 1 thinking, the accuracy of their decisions might be compromised. Thus, System 2 thinking might find the faults in simple System 1 judgment [39]. However, the two can complement each other. If information can be easily and fluently processed, System 1 is triggered [40]. When individuals face difficulty processing information, or their hasty decisions could be faulty, System 2 is preferred for guiding their judgment [41]. In conclusion, information processing strategies are determined by the ease or difficulty with which information is processed and the cognitive demands of the task at hand [42].

Live and static comments—due to their distinct characteristics—may engage different cognitive systems in users. Specifically, the dynamic nature of live comments is likely to promote System 1 thinking, encouraging intuitive and automatic information processing. In contrast, the fixed and asynchronous format of static comments provides opportunities for deeper reflection, facilitating System 2 processing. These differences in cognitive engagement may result in varied ways in which users evaluate video content depending on the type of comment they are exposed to. Below, we propose hypotheses based on dual system theory.

2.3. Effect of Live Comments on User Evaluation of Online Videos

Live comments are real-time interactions that enable users to express their emotions or viewpoints regarding specific moments in a video as they watch it. These comments appear synchronously on the upper layer of the video content at the corresponding timestamp [4,27]. Compared to static comments, live comments are generally shorter and more fragmented, with lower linguistic complexity. However, they offer near-instantaneous feedback and are temporally aligned with the unfolding video content. This trade-off between depth and immediacy allows live comments to capture users' spontaneous emotional reactions. Information richness is commonly evaluated across four dimensions: feedback, multiple cues, language variety, and personal focus [43]. Static comments tend to perform better on these dimensions, except feedback, indicating that live comments carry relatively lower information richness. This characteristic enables users to grasp the message without engaging in deep cognitive processing. Consequently, users are more likely to rely on intuitive System 1 processing when interpreting live comments.

Furthermore, research indicates that the emotional tendencies represented in live comments, which align with the video content in real time, exhibit a pronounced herding effect [29]. According to the theory of emotional contagion, individuals tend to unconsciously mimic and internalize the emotional expressions of those around them [44]. Consequently,

the emotional contagion triggered by the herding effect enables users to resonate strongly with the emotions expressed in live comments, thereby enhancing their overall engagement and emotional connection with the content [45]. As users navigate the emotional landscape conveyed by live comments within the constraints of time, they are inclined to rely more on System 1 thinking for subsequent decisions, such as expressing appreciation [46]. Building on the concept that System 1 is fast, intuitive, and mostly emotion driven [47,48], we hypothesize that a higher prevalence of negative emotions in live comments will intensify immediate negative emotional perceptions and decrease users' likelihood of providing favorable evaluations of the video. Conversely, a higher proportion of positive emotions in live comments is expected to promote favorable perceptions of the video, thereby increasing users' likelihood of providing favorable evaluations. Therefore, we propose the following hypotheses:

Hypothesis 1a. *The proportion of positive live comments has a significant positive effect on user evaluation of online videos.*

Hypothesis 1b. *The proportion of negative live comments has a significant negative effect on user evaluation of online videos.*

2.4. Effect of Static Comments on User Evaluation of Online Videos

In contrast to live comments, static comments are typically displayed in a fixed area and appear concurrently with the video content, leading to a diversion of users' attention from the video to the comments section [49]. Moreover, static comments allow for multi-paragraph expressions, the use of multimodal elements (e.g., emojis, images), and layered reasoning, resulting in greater linguistic and cue diversity compared to live comments—and thus, higher informational richness. Also, users often seek out static comments to evaluate the opinions of others regarding the video. Consequently, compared to the immediate emotional engagement elicited by live comments, users' focus has shifted from the video content to static comments, either upon completing the video or during pauses. When exposed to static comments with higher information richness and no time constraints on information processing, users are more likely to engage in System 2 thinking. This enables them to process information more deliberately and rationally before making decisions [48]. This cognitive approach involves careful reading and a thorough digestion of information, resulting in a slower and more considered decision-making process

Additionally, static comments often serve as a direct reflection of the ideas and opinions presented in the video [50]. Although emotional contagion may still occur, it tends to proceed at a slower pace because users have already exited the immersive viewing experience. As such, if these comments predominantly exhibit positive sentiment, users can infer that the favorable discourse arises from the video's content, leading to greater appreciation for the creator's efforts. Conversely, when negative sentiment predominates in static comments, users may interpret a correlation between these remarks and unfavorable aspects of the video. Prior research has shown that presenting both sides of an issue can significantly enhance message credibility [51]. Accordingly, this candid presentation of both positive and negative viewpoints may foster a perception among users that the vlogger is objective and trustworthy, thereby increasing their willingness to express appreciation as a testament to the vlogger's balanced content creation approach. Therefore, we propose the following hypotheses:

Hypothesis 2a. *The proportion of positive static comments has a significantly positive effect on user evaluation of online videos.*

Hypothesis 2b. *The proportion of negative static comments has a significantly positive effect on user evaluation of online videos.*

2.5. Moderating Effect of Video Type

The conceptual framework of value perception posits that users evaluate online services along two primary dimensions: informational value and emotional value [52,53]. Accordingly, we categorize videos into two types: those with elevated informational value (i.e., information-type videos) and those with heightened emotional value (i.e., emotional-type videos). Informational-type videos address users' cognitive needs by providing substantial information and facilitating rational discussions about the video content or the vlogger [19]. In contrast, emotional-type videos predominantly evoke positive emotions to capture user attention, acknowledging that viewers often seek videos for relaxation, happiness, and entertainment [54].

Given the high engagement and brief presence of live comments during video content consumption, users must be highly focused to process information rapidly. This immersion may induce a flow state, illustrating the optimization and efficacy of live comments [55]. Consequently, users are likely to engage primarily in System 1 thinking rather than System 2 when processing information from live comments, regardless of the video type. Essentially, users may exhibit consistent thinking and information processing patterns when interacting with live comments across various video types.

However, the effect of static comments may vary between information-type and emotional-type videos. As previously mentioned, users typically engage System 2 thinking to comprehend and assimilate information from static comments. For information-type videos, which exhibit minimal emotional fluctuations, viewers often emerge from the viewing experience in a calmer and more rational state. This environment facilitates the activation of System 2 thinking, allowing for thorough information processing and independent decision-making. However, emotional-type videos, laden with emotions and valence, can impose lingering aftereffects on users, even after the videos have concluded. The nature and duration of these aftereffects depend on the characteristics of the stimuli as well as individual psychological traits. For instance, video attributes such as duration and content consistency align more closely with the stimulus properties that evoke assimilative aftereffects [56–58]. Consequently, users may experience a prolonged period during which these aftereffects potentially undermine their ability to engage in System 2 thinking. Therefore, we propose that the positive effect of both positive and negative static comments on user evaluation of online videos will be enhanced in the context of information-type videos. The following hypotheses are proposed:

Hypothesis 3a. *The positive effect of negative static comments on user evaluation of information-type videos is stronger than that of emotional-type videos.*

Hypothesis 3b. *The positive effect of positive static comments on user evaluation of information-type videos is stronger than that of emotional-type videos.*

2.6. Moderating Effect of Health Threats

Significant health threats elicit a range of emotional responses, including depression, anxiety, anger, and other mental disorders, which can provoke physiological reactions [59,60]. When individuals perceive a decline in their sense of health control, they often experience an increased desire to restore this control [61]. In situations where they cannot eliminate or avoid the source of health threats, individuals might adopt behaviors aimed at managing their emotions to achieve emotional release [62]. Such behaviors

may involve engaging in distracting and enjoyable activities or seeking satisfying support to cope with and enhance their emotional well-being during these challenging times [63].

To mitigate the negative effects of health threats, individuals have shifted various offline consumption activities to online platforms [64]. Watching movies, for example, is a means of distraction and pleasure enhancement that can offer relaxation and entertainment [54,63]. The features of self-disclosure and the social interaction feature of video platforms can alleviate the emotional stress induced by health threats and address individuals' social needs.

Emotional regulation theory posits that individuals in negative emotional states often actively seek to repair and regulate their emotions through proactive behaviors [65]. As noted in 2.3, the herding effect generated by the emotional tendencies of live comments may allow users to experience similar emotions. In this context, positive live comments demonstrate a more pronounced positive effect, particularly given the heightened demand for alleviating negative emotional states during health threats. This positive effect is further intensified, fulfilling users' emotional regulation needs and increasing their likelihood of expressing favorable evaluations of the content they consume. Therefore, during health threats, the positive effect of positive live comments on users' evaluations of online videos is expected to be stronger. Conversely, users tend to avoid negative live comments during this period, indicating that negative comments do not exert a significant influence.

By contrast, when processing information from static comments, users typically engage in System 2 thinking, resulting in comparatively smaller emotional fluctuations. We thus anticipate that static comments are less affected by health threats than live comments. Accordingly, we propose the following hypotheses:

Hypothesis 4. *During health threats, the proportion of positive live comments has a more positive effect on user evaluation of online videos.*

3. Data and Method

3.1. Data Collection

We focus on videos posted on the Bilibili platform, a popular Chinese video-sharing website. According to Bilibili's first-quarter report for 2020, the platform had about 50 million active daily users, 1.1 billion daily video plays, and 4.9 billion interactions per month, indicating a high level of overall user engagement.

In July 2020, we started to use web-crawling tools to collect all videos from vloggers listed in Bilibili's "Top 100" to reduce interference from low-interaction, low-traffic videos. Additionally, we categorized the video types based on the tags associated with the collected content to differentiate between informational and emotional categories. For example, a "knowledge" tag often indicates that the video aims to convey information, whereas a "comedy" tag usually shows that the video predominantly elicits positive emotions. After excluding outliers such as repeated videos and those with fewer than 20 views, the dataset comprises 667 videos, 1.59 million static comments, and 570,000 live comments. Excluding videos with under 20 views ensures that each video retained a sufficient volume of live and static comment data for robust analysis.

In addition to collecting data for static and live comments, we gather information about the number of coins rewarded for each video as the dependent variable. Coins are a form of virtual currency users earn through daily login and video sharing, with an upper limit on coin acquisition set by the platform (e.g., two coins per day). Due to the limited availability of coins, users carefully consider whether a video is worth spending their valuable coins for evaluation when deciding to reward it. Studies have recognized coin-rewarding behavior as an effective indicator of audience appreciation [4]. Therefore,

we use coin-rewarding behavior as a quantitative measure to assess user evaluations of online videos. Control variables include the number of views, shares, likes, and vlogger fans, as well as video length.

For the sentiment analysis of static comments and live comments, we use the pre-trained ERNIE-SETA text analysis tool. ERNIE-SETA leverages unsupervised training and integrates emotional knowledge into its model to enhance the accuracy of emotional analysis results. It has demonstrated strong performance across multiple datasets, achieving a classification accuracy of 97.6% on the SST-2 English dataset and 96.5% on the ChnSentiCorp Chinese dataset [66]. This approach aligns with the methodologies employed in several social media-related studies that have effectively utilized ERNIE-SETA for sentiment analysis, demonstrating its robustness in capturing the nuances of public sentiment across various online platforms [67,68]. The ERNIE-SETA text analysis tool provides not only definitive judgment outcomes but also confidence level for these results. In our approach, texts with a confidence level exceeding 90% for either positive or negative sentiments are considered as conclusive judgments. Conversely, texts that fall below this threshold for both sentiment categories are categorized as neutral. This selective adoption of high-confidence predictions aims to enhance the model's performance by minimizing uncertainty in sentiment classification. To address potential biases arising from varying interaction levels among different videos and emotional polarity, we use the proportion of positive or negative texts to the totality of texts with positive or negative emotional polarity as the independent variable for subsequent analysis.

To examine the impact of health threats, this study uses the COVID-19 pandemic as a representative variable. The outbreak, which began in 2020, significantly disrupted daily life and serves as a critical case for studying public health crises. Videos are categorized into pre-pandemic and pandemic periods based on their upload dates, with 23 January 2020 serving as the demarcation line. Additionally, timestamps for live comments and static comments are filtered to align with the respective periods. Specifically, for videos posted before the pandemic, any comments made during the pandemic period have been excluded from the analysis. Furthermore, given that videos uploaded close to the cutoff date could have comments distributed across both time frames, data from three days before and after 23 January 2020 have been omitted to ensure a clear distinction between the two periods.

3.2. Variable Definition

Table 1 summarizes the variables used in this study. The dependent variable Coin_i is the number of coins for the i-th video, while the independent variables are the proportions of positive and negative live comments, as well as the proportions of positive and negative static comments, each calculated as the proportion of comments with a specific sentiment (positive or negative) for the i-th video out of all comments with the same sentiment polarity. To alleviate the influence of heteroscedasticity, we log-transform both the independent and dependent variables. Moreover, we utilize the occurrence of COVID-19 at the time of video release as an indicator of health threats, with this factor and the types of videos serving as moderating variables. Specifically, Pandemic_i indicates whether the video was released during the pandemic. Informational_i reflects whether it is an informational video. Furthermore, to mitigate the endogeneity issues stemming from unobserved vlogger characteristics (e.g., creative style and brand influence), we incorporate vlogger fixed effects in our regression by including a dummy variable for each vlogger using the i-th vlogger's personal code Uploder_Code_i. Considering that a creator's reputation and fan engagement may affect fan interactions, we use the number of the i-th vlogger's followers, Follower_i, as a proxy for reputation. The follower count captures the platform algorithm's effect in boosting a video's exposure, thereby reducing the risk of omitting important variables. We

also use the i-th vlogger's personal code $Uploder_Code_i$ as a proxy variable for personal characteristics. In addition, we select the i-th video length $Video_Length_i$, the number of the i-th vlogger's followers $Follower_i$, the number of likes $Video_Like_i$ for the i-th video, the number of shares $Video_Share_i$ for the i-th video, and the number of views $Video_View_i$ for the i-th video as control variables.

Table 1. Definition of variables.

Variable	Definition
$Coin_i$	Number of coins for the i-th video
$Positive_Live_i$	Proportion of positive live comments out of all comments with positive emotional polarity of the i-th video
$Negative_Live_i$	Proportion of negative live comments out of all comments with negative emotional polarity of the i-th video
$Positive_Static_i$	Proportion of positive static comments out of all comments with positive emotional polarity of the i-th video
$Negative_Static_i$	Proportion of negative static comments out of all comments with negative emotional polarity of the i-th video
$Pandemic_i$	Dummy variable indicating whether the focal video was released during the pandemic
$Informational_i$	Dummy variable indicating whether the focal video was emotional
$Follower_i$	Number of followers for the i-th vlogger
$Video_Length_i$	Length of the i-th video
$Video_View_i$	Number of views for the i-th video
$Video_Like_i$	Number of likes for the i-th video
$Video_Share_i$	Number of shares for the i-th video
$Uploader_Code_i$	The i-th vlogger's personal code

4. Empirical Results

4.1. Study 1: Effect of Live Comments and Static Comments on Coin-Rewarding Behavior

We conducted an OLS robust regression analysis of the video data using STATA 14. As shown in Table 2, the proportion of positive live comments ($\beta = 0.300$, $p < 0.01$) exhibits a significantly positive effect, providing support for H1b. This finding implies that a higher proportion of positive live comments corresponds to an increased likelihood of triggering coin-rewarding behavior among users. However, the proportion of negative live comments has no significant effect ($\beta = -0.076$, $p > 0.05$). This could be attributable to the heightened effect of negative emotions during the pandemic, leading to a weakened negative influence on people's coin-rewarding behavior. Thus, H1a is not supported.

With respect to static comments, the results presented in Table 2 reveal significant positive effects for both negative and positive static comments. Specifically, a higher proportion of negative static comments exhibits a significantly positive effect on coin-rewarding behavior ($\beta = 0.801$, $p < 0.001$), and similarly, a higher proportion of positive static comments also predicts more rewards ($\beta = 0.825$, $p < 0.001$). These findings provide joint support for H2a and H2b, suggesting that static comments—regardless of whether they convey praise or criticism—are associated with an increased likelihood of viewers rewarding content with coins. Furthermore, this observation validates that the presentation of both positive and negative information can significantly enhance users' trust in the information source [51], thereby making them more willing to reward coins to the video.

Table 2. Results of the effect of live comments and static comments on coin-rewarding behavior.

Variables	ln $Coin_i$
Main Effects	
ln $Negative_Live_i$	−0.076 (0.089)
ln $Negative_Static_i$	0.801 *** (0.113)
ln $Positive_Live_i$	0.300 ** (0.110)
ln $Positive_Static_i$	0.825 *** (0.158)
Control Variables	
$Informational_i$	−0.415 *** (0.084)
$Follower_i$	1.23×10^{-7} *** (3.25×10^{-8})
$Video_Length_i$	−0.0001 ** (0.00003)
$Video_view_i$	2.48×10^{-7} ** (7.39×10^{-8})
$Video_Like_i$	0.00002 ** (7.27×10^{-6})
$Video_Share_i$	-5.67×10^{-6} (4.10×10^{-6})
$Pandemic_i$	Control
$Uploader_Code_i$	Control
Model Evaluation	
R^2	0.742
Observations	667

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, standard errors in parentheses ().

4.2. Study 2: Moderating Effect of Video Type

Building on the model in Study 1, we incorporate the classification of informational and emotional videos as moderator variables to investigate potential divergences in the effect of live comments and static comments on coin-rewarding behavior across different video types. Table 3 shows the results. Model 2 indicates a significant positive moderating effect of video types on the effect of negative static comments on coin-rewarding behavior ($\beta = 0.604$, $p < 0.05$). This suggests that the positive effect of negative static comments on coin-rewarding behavior for informational videos is stronger than that for emotional videos. This may be because the primary goal of informational videos is to convey knowledge or facts, which tend to evoke less emotional arousal. In such cases, viewers are more likely to engage in systematic, analytical thinking (System 2 processing), leading them to evaluate static comments more thoughtfully and rationally. As a result, the perceived credibility of information resulting from the dual presentation of content is more pronounced in the context of informational videos.

However, we find no significant moderating effect of video types on the effect of positive static comments on coin-rewarding behavior ($\beta = -0.240$, $p > 0.05$). This could be attributable to the ceiling effect generated by positive static comments. Once this positive effect reaches its peak, it becomes unchanged for both information-type and emotional-type videos. Additionally, we find that the moderating effect of video type on live comments is not significant, aligning with our expectation. These results support H3a and reject H3b.

Table 3. Results of moderating effect of video type.

Variable	ln $Coin_i$	
	Model 1	**Model 2**
Main Effect		
ln $Negative_Live_i$	−0.076 (0.089)	−0.150 (0.102)
ln $Negative_Static_i$	0.801 *** (0.113)	0.632 *** (0.121)
ln $Positive_Live_i$	0.300 ** (0.110)	0.459 ** (0.153)
ln $Positive_Static_i$	0.825 *** (0.158)	1.023 *** (0.173)
$Informational_i$	−0.415 *** (0.084)	−0.487 *** (0.088)
Interaction Term		
$Informational_i$ × ln $Negative_Live_i$		0.089 (0.219)
$Informational_i$ × ln $Negative_Static_i$		0.604 * (0.237)
$Informational_i$ × ln $Positive_Live_i$		0.284 (0.325)
$Informational_i$ × ln $Positive_Static_i$		−0.240 (0.351)
Control Variable		
$Follower_i$	1.23×10^{-7} *** (3.25×10^{-8})	1.07×10^{-7} ** (3.34×10^{-8})
$Video_Length_i$	−0.0001 ** (0.00003)	-0.0001 *** (0.00003)
$Video_View_i$	2.48×10^{-7} ** (7.39×10^{-8})	2.62×10^{-7} *** (7.48×10^{-8})
$Video_Like_i$	0.00002 ** (7.27×10^{-6})	0.00002 ** (7.13×10^{-6})
$Video_Share_i$	-5.67×10^{-6} (4.10×10^{-6})	-5.22×10^{-6} (4.31×10^{-6})
$Uploader_Code_i$	Control	Control
$Pandemic_i$	Control	Control
Model Evaluation		
R^2	0.742	0.746
Observation	667	667

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, standard errors in parentheses ().

4.3. Study 3: Moderating Effect of Health Threats

We use COVID-19 as a moderating variable in the model to explore whether the effects of live comments and static comments on the number of coins differ during various health threat periods. Table 4 shows the results. In Model 2, there is a significant positive moderating effect of COVID-19 on the effect of positive live comments on coin-rewarding behavior ($\beta = 0.713$, $p < 0.05$). This suggests that the effect of positive live comments on coin-rewarding behavior during the COVID-19 pandemic is stronger than that before the pandemic. This finding aligns with prior research suggesting that individuals tend to seek out positive information to regulate their emotions and counteract the negative emotional states induced by health threats [63]. However, the model shows no significant moderating effect of COVID-19 on the effect of negative live comments on coin-rewarding behavior ($\beta = -0.210$, $p > 0.05$), suggesting that the impact of negative live comments remained stable despite the heightened negative emotional context caused by health threats. Additionally,

the moderating effect of the COVID-19 pandemic on static comments was found to be nonsignificant. Consequently, H4 is supported. These findings further clarify the boundary conditions under which live comments influence user behavior in evaluating videos.

Table 4. Results of moderating effect of health threats.

Variable	ln $Coin_i$	
	Model 1	**Model 2**
Main Effect		
ln $Negative_Live_i$	−0.076 (0.089)	0.035 (0.095)
ln $Negative_Static_i$	0.801 *** (0.113)	0.849 *** (0.111)
ln $Positive_Live_i$	0.300 ** (0.110)	0.245 * (0.106)
ln $Positive_Static_i$	0.825 *** (0.158)	0.790 *** (0.152)
$Pandemic_i$	0.235 ** (0.079)	0.208 ** (0.076)
Interaction Term		
$Pandemic_i$ × ln $Negative_Live_i$		−0.210 (0.205)
$Pandemic_i$ × ln $Negative_Static_i$		−0.058 (0.222)
$Pandemic_i$ × ln $Positive_Live_i$		0.713 ** (0.237)
$Pandemic_i$ × ln $Positive_Static_i$		0.584 (0.330)
Control Variable		
$Follower_i$	1.23×10^{-7} *** (3.25×10^{-8})	1.02×10^{-7} ** (3.23×10^{-8})
$Video_Length_i$	−0.0001 ** (0.00003)	−0.0001 ** (0.00003)
$Video_View_i$	2.48×10^{-7} ** (7.39×10^{-8})	3.12×10^{-7} *** (7.70×10^{-8})
$Video_Like_i$	0.00002 ** (7.27×10^{-6})	0.00002 ** (7.27×10^{-6})
$Video_Share_i$	-5.67×10^{-6} (4.10×10^{-6})	-5.91×10^{-6} (4.08×10^{-6})
$Uploader_Code_i$	Control	Control
$Informational_i$	Control	Control
Model Evaluation		
R^2	0.742	0.747
Observation	667	667

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, standard errors in parentheses ().

4.4. Robustness Check

To further verify the robustness of the above findings, we conducted three robustness checks. Firstly, to examine the moderating effects and other potential interactive effects, we add the pandemic and video type variables as moderators to the same regression model. The analysis reveals significant positive effects on users' coin-rewarding behavior from both live and static positive comments, as well as from negative static comments. Notably, the influence of positive live comments on this behavior persists significantly even when both moderating variables are included in the model simultaneously. The impact of positive live comments on user engagement through coin-rewarding has intensified during the pandemic period (β = 0.637, $p < 0.05$), demonstrating a stronger effect compared

to pre-pandemic levels. In addition, negative static comments exhibit distinct impacts on coin-rewarding behavior depending on whether the video content is informational or emotional. Specifically, negative static comments have a more pronounced positive effect on coin-rewarding behavior in informational videos relative to emotional ones ($\beta = 0.551$, $p < 0.05$). These results show a pattern consistent with the main model and previous data analysis. The full results reported in Appendix A Table A1 confirm the stability of our key findings.

Secondly, although the number of coins serves as a strong proxy for user engagement with videos [4], we also considered likes—a commonly used measure of user behavior in social media research [69,70]—as an alternative dependent variable. We reran the main and moderating effect analyses using the number of likes as the outcome variable. The results are summarized in Table 5. Model 1 shows that both positive and negative static comments, along with positive live comments, significantly increase the number of likes, whereas negative live comments do not yield a significant effect. This is consistent with our earlier findings. Models 2 and 3 test the moderating roles of health threat and video type, respectively. Overall, these results largely align with our previous analyses. However, in this context, the interaction between health threat and positive static comments becomes significant ($\beta = 0.531$, $p < 0.05$), indicating that during the pandemic, positive static comments—similar to live comments—may enhance users' liking behavior by influencing emotional states.

Table 5. Robustness tests with alternative engagement metrics.

Variable	ln Like$_i$		
	Model 1	**Model 2**	**Model 3**
Main Effect			
ln Negative_Live$_i$	0.033 (0.083)	0.101 (0.090)	−0.009 (0.097)
ln Negative_Static$_i$	0.720 *** (0.092)	0.767 *** (0.093)	0.627 *** (0.096)
ln Positive_Live$_i$	0.210 * (0.096)	0.185 (0.099)	0.335 ** (0.127)
ln Positive_Static$_i$	0.744 *** (0.121)	0.728 *** (0.116)	0.882 *** (0.133)
Pandemic$_i$	0.260 *** (0.065)	0.255 *** (0.064)	0.252 *** (0.065)
Informational$_i$	0.116 (0.073)	0.115 (0.072)	0.057 (0.078)
Interaction Term			
Pandemic$_i$ × ln Negative_Live$_i$		−0.013 (0.204)	
Pandemic$_i$ × ln Negative_Static$_i$		0.240 (0.188)	
Pandemic$_i$ × ln Positive_Live$_i$		0.504 * (0.228)	
Pandemic$_i$ × ln Positive_Static$_i$		0.531 * (0.257)	
Informational$_i$ × ln Negative_Live$_i$			0.097 (0.205)
Informational$_i$ × ln Negative_Static$_i$			0.375 * (0.186)
Informational$_i$ × ln Positive_Live$_i$			0.242 (0.272)
Informational$_i$ × ln Positive_Static$_i$			−0.068 (0.277)

Table 5. *Cont.*

Variable	ln $Like_i$		
	Model 1	**Model 2**	**Model 3**
Control Variable			
$Follower_i$	4.06×10^{-8} (2.77×10^{-8})	2.63×10^{-8} (2.59×10^{-8})	2.71×10^{-8} (2.62×10^{-8})
$Video_Length_i$	−0.00005 * (0.00002)	−0.00004 * (0.00002)	−0.00004 ** (0.00002)
$Video_View_i$	3.37×10^{-7} *** (4.89×10^{-8})	3.72×10^{-7} *** (4.51×10^{-8})	3.52×10^{-7} *** (4.52×10^{-8})
$Video_Coin_i$	6.08×10^{-6} *** (7.27×10^{-7})	5.97×10^{-6} *** (7.23×10^{-7})	6.11×10^{-6} *** (7.37×10^{-7})
$Video_Share_i$	-3.09×10^{-6} (2.19×10^{-6})	-3.18×10^{-6} (2.07×10^{-6})	-2.50×10^{-6} (2.37×10^{-6})
$Uploader_Code_i$	Control	Control	Control
Model Evaluation			
R^2	0.736	0.742	0.739
Observation	667	667	667

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, standard errors in parentheses ().

Finally, given the potential issues of violations of distributional assumptions in OLS models when applied to discrete count data, we conducted a sensitivity analysis using Negative Binomial regression. The results show that while some significance levels differ slightly, the overall findings remain largely consistent with our previous analyses. Additionally, like the analysis using the alternative dependent variable, the interaction term between health threat and positive static comments remains significant (β = 0.768, $p < 0.05$). The full results are reported in Appendix A Table A2.

Overall, all three robustness checks align closely with the main analysis. However, in some of the moderation analyses, the positive effect of positive static comments on users' coin-rewarding and like behaviors appears to be stronger during periods of heightened health threat. This may be because, although users generally rely more on System 2 processing when reading static comments—due to the lack of real-time constraints and richer linguistic and contextual cues—during the pandemic, users may experience anxiety or other negative emotions due to health-related concerns, which enhances the emotion-regulating function of positive static comments. Limited by the differences in the distributional assumptions across regression models, this moderating effect was not fully captured in the main analysis, where it only showed marginal significance (β = 0.584, $p = 0.08$).

5. Discussion

5.1. Research Results

This study uses natural language processing (NLP) text analysis tools to investigate two types of interactive comments—live comments and static comments—in videos across two distinct categories: emotional videos and informational videos. We aim to examine how these interactive texts influence user evaluation of online videos in different ways.

The findings reveal, first, that live comments, as a novel format characterized by strong real-time interactions, provide a platform for users to express their emotions both intensely and extensively. Videos featuring a higher proportion of positive live comments tend to enhance users' willingness to reward coins. In contrast, videos with a higher proportion of negative live comments do not have a significant impact on users' willingness to reward coins. However, we observe that both negative and positive static comments exert a significant positive effect on users' willingness to reward coins.

Our first moderation analysis indicates that observed differences in the effects of live and static comments can be attributed to the type of video. Notably, negative static comments in informational videos have a more substantial impact on enhancing users' willingness to reward coins. The effect of positive static comments remains consistent across both informational and emotional videos. This result confirms that live comments are not influenced by the inherent characteristics of the video. Furthermore, the impact of static comments with different emotional tendencies on users' coin-rewarding behavior differs significantly by video type. This suggests that the dual presentation of information (both positive and negative) enhances the perceived credibility of information sources in informational videos to a greater extent, whereas positive static comments may exhibit a ceiling effect, limiting observable differences across video types.

Finally, we introduce health threats as a moderator to distinguish the effects between live comments and static comments. The proportion of positive live comments exhibits a stronger positive effect on users' willingness to reward coins during health threats. Thus, health threats amplify the influence of positive live comments on users' propensity to reward coins. Robustness checks further reveal that when using the number of likes as the dependent variable, as well as when applying a Negative Binomial regression model, positive static comments also exert a stronger positive influence on user coin-rewarding behavior during the pandemic.

5.2. Theoretical Contribution

This study contributes to information processing research and introduces a novel application of dual system theory, thus enriching the digital marketing literature. Existing studies suggest that under time constraints, individuals tend to rely on quick, intuitive processing, while they engage in more deliberate, cognitive processing when time is not constrained [71]. By analyzing how the emotional polarity of live and static comments—under varying time constraints and different levels of information richness—affects user engagement, we found that users respond differently to live and static comments with the same emotional polarity. This observation provides empirical evidence for differentiated information processing pathways depending on time constraints and information richness. Our findings support theoretical expectations and offer a cognitive perspective on information processing pathways within the dual system theory framework, particularly in contexts with varying time limitations.

Moreover, our study identifies two key boundary conditions in the processing of live versus static comments. First, negative comments have a more pronounced positive impact on user evaluations of informational videos compared to emotional videos. Informational videos aim to fulfill users' informational needs [19], and in non-time-constrained contexts, individuals are more likely to engage in rational and deliberate cognitive processing to meet these needs [17,72]. As a result, users may prioritize the content's informational value over its emotional tone when they encounter negative comments in informational videos. Previous research indicates that users perceive negative comments as more useful [73], which, while potentially decreasing purchase intent, enhances appreciation of the video's content. This aligns with the concept of negativity bias, where negative information garners more attention [74], offering empirical evidence of the differentiated impact of negative comments on video creators and brands. Additionally, we observed that positive live comments have a stronger positive effect on user engagement during the pandemic than during non-pandemic periods. This suggests that, in times of health threats, users turn to online video platforms more for emotional regulation than for information alone. Robustness checks further revealed that positive static comments also produce similar emotional regulation effects during the pandemic, thereby encouraging users to reward

coins. These results underscore the important role that positively valenced commentary plays in influencing user behavior during health threats.

5.3. Managerial Implications

The influence of live and static comments on user evaluation of online videos represents a critical consideration for both video platforms and vloggers. Firstly, vloggers should clearly differentiate the roles of live and static comments in emotional regulation versus information dissemination. Vloggers should employ tailored strategies to address the varying types of expressions. Furthermore, they should develop video production plans specifically designed for the types of videos they specialize in. For informational videos, the emphasis should be on addressing the informational value, particularly when handling negative static comments. Vloggers should monitor whether negative comments are directed at the content or themselves and consider adjusting the tone of their responses accordingly. For emotional videos, vloggers should focus on conveying positive emotional value to generate more positive live and static comments, which in turn encourage increased coin rewarding behavior among users.

Secondly, during periods of health threats, vloggers should place more emphasis on managing emotional responses through live comments. By integrating emotional cues in real-time (e.g., using more empathetic or comforting language), vloggers can engage users' emotional needs and enhance user satisfaction.

Thirdly, the emotional polarity of live comments offers an immediate gauge of how users perceive a brand in the moment. Marketers should actively monitor live comments for real-time feedback, using these data to adjust marketing strategies quickly and effectively, especially during periods of heightened public emotion, such as health crises or social unrest. While live comments capture users' immediate emotional responses, static comments offer deeper reflections. Brands should adopt a dual-analysis approach, where live comments are used to measure immediate reactions to promotional campaigns or brand messaging, while static comments are analyzed for long-term perceptions and deeper insights into customer needs.

Lastly, platforms should enhance features that facilitate real-time interaction through live comments, providing users with more emotional regulation tools, such as customizable emotional responses (e.g., emojis or quick emotional feedback buttons). These features should be designed with the awareness that live comments are often more effective during health threats and other stressful periods. Additionally, due to screen size limitations that allow only a selection of live comments to be displayed, platforms should develop more targeted algorithms. These algorithms would increase the display frequency of positive live comments, further enhancing their emotional regulation effect during health threats.

6. Limitations and Future Research

While this study provides some valuable insights into video platform user behavior and its underlying mechanisms, several limitations should be noted. Firstly, our reliance on online data introduces a risk of selection bias: our sample predominantly comprises popular videos and highly active users, potentially excluding fewer active participants and less-viewed content. Although we made every effort to lower the view count threshold to 20, this selection bias may still reduce the representativeness of our data. In addition, although online data allow for a broad collection and the use of measures to mitigate potential endogeneity (e.g., including blogger IDs as fixed effects in the regression models), inherent data structure limitations may still introduce bias. Therefore, future research should consider obtaining panel data through long-term tracking to avoid potential endogeneity problems. Moreover, future research might further investigate potential omitted variables

and extend the dataset to include videos from vloggers beyond the "Top 100", thereby capturing a broader range of video features and mitigating potential selection bias.

Secondly, the cultural context of Bilibili poses another significant constraint. The platform's distinct subcultural characteristics and its predominantly young user base mean that the results derived may not easily transfer to other social video platforms like YouTube or TikTok, which operate under different cultural norms and demographic profiles. Future research should explore these cultural nuances and examine the robustness of the findings in other contexts.

Furthermore, the relatively short time span of data collection hampers a comprehensive analysis of temporal dynamics, particularly regarding daily behavioral variations. Future studies could extend the data collection period and employ upgraded technical tools to construct more nuanced panel data, thereby capturing both long-term trends and short-term fluctuations.

Lastly, the narrow selection of video types examined in this study limits the scope of our analysis. Broadening the range of content categories in future research would help determine whether the observed patterns hold across diverse forms of video content. To address these limitations, future research may benefit from incorporating multifaceted psychological experiments and more robust experimental or quasi-experimental designs to validate and refine the theoretical framework proposed in this study.

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Appendix A

Table A1. Results of regression analysis with combined moderators.

Variable	ln $Coin_i$
Main Effect	
ln $Negative_Live_i$	−0.053 (0.106)
ln $Negative_Static_i$	0.681 *** (0.120)
ln $Positive_Live_i$	0.406 ** (0.148)
ln $Positive_Static_i$	0.981 *** (0.165)
$Pandemic_i$	0.199 * (0.077)
$Informational_i$	−0.480 *** (0.086)

Table A1. *Cont.*

Variable	ln $Coin_i$
Interaction Term	
$Pandemic_i$ × ln $Negative_Live_i$	−0.160 (0.212)
$Pandemic_i$ × ln $Negative_Static_i$	−0.072 (0.236)
$Pandemic_i$ × ln $Positive_Live_i$	0.637 ** (0.241)
$Pandemic_i$ × ln $Positive_Static_i$	0.571 (0.332)
$Pandemic_i$ × $Informational_i$	−0.011 (0.170)
$Informational_i$ × ln $Negative_Live_i$	0.049 (0.213)
$Informational_i$ × ln $Negative_Static_i$	0.551 * (0.235)
$Informational_i$ × ln $Positive_Live_i$	0.288 (0.308)
$Informational_i$ × ln $Positive_Static_i$	−0.209 (0.337)
Control Variable	
$Follower_i$	8.85×10^{-8} * (3.40×10^{-8})
$Video_Length_i$	−0.0001 ** (0.00003)
$Video_View_i$	3.21×10^{-7} *** (7.90×10^{-8})
$Video_Like_i$	0.00002 ** (7.15×10^{-6})
$Video_Share_i$	-5.53×10^{-6} (4.31×10^{-6})
$Uploader_Code_i$	Control
Model Evaluation	
R^2	0.750
Observation	667

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, standard errors in parentheses ().

Table A2. Consistency across estimation methods.

Variable	ln $Coin_i$		
	Model 1	**Model 2**	**Model 3**
Main Effect			
ln $Negative_Live_i$	−0.010 (0.070)	0.110 (0.071)	−0.089 (0.087)
ln $Negative_Static_i$	0.765 *** (0.096)	0.817 *** (0.092)	0.554 *** (0.113)
ln $Positive_Live_i$	0.190 * (0.093)	0.160 (0.087)	0.296 * (0.148)
ln $Positive_Static_i$	0.576 ** (0.174)	0.607 *** (0.145)	0.792 *** (0.191)
$Pandemic_i$	0.142 * (0.071)	0.124 (0.067)	0.128 (0.069)
$Informational_i$	−0.383 *** (0.076)	−0.377 *** (0.076)	−0.412 *** (0.076)

Table A2. *Cont.*

Variable	ln $Coin_i$		
	Model 1	**Model 2**	**Model 3**
Interaction Term			
$Pandemic_i \times$ ln $Negative_Live_i$		−0.215 (0.158)	
$Pandemic_i \times$ ln $Negative_Static_i$		−0.072 (0.176)	
$Pandemic_i \times$ ln $Positive_Live_i$		0.735 *** (0.196)	
$Pandemic_i \times$ ln $Positive_Static_i$		0.768 * (0.315)	
$Informational_i \times$ ln $Negative_Live_i$			0.054 (0.190)
$Informational_i \times$ ln $Negative_Static_i$			0.632 ** (0.214)
$Informational_i \times$ ln $Positive_Live_i$			0.064 (0.332)
$Informational_i \times$ ln $Positive_Static_i$			−0.532 (0.410)
Control Variable			
$Follower_i$	9.44×10^{-8} *** (2.62×10^{-8})	7.54×10^{-8} ** (2.50×10^{-8})	8.50×10^{-8} ** (2.54×10^{-8})
$Video_Length_i$	-0.00008 (0.00004)	−0.00008 * (0.00003)	−0.00008 * (0.00004)
$Video_View_i$	1.39×10^{-7} * (5.98×10^{-8})	2.09×10^{-7} ** (6.05×10^{-8})	1.52×10^{-7} ** (5.67×10^{-8})
$Video_Like_i$	0.00004 *** (5.55×10^{-6})	0.00004 *** (5.45×10^{-6})	0.00004 *** (5.61×10^{-6})
$Video_Share_i$	−0.00001 * (4.78×10^{-6})	−0.00001 ** (4.05×10^{-6})	−0.00001 * (5.00×10^{-6})
$Uploader_Code_i$	Control	Control	Control
Model Evaluation			
R^2	0.071	0.072	0.072
Observation	667	667	667

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, standard errors in parentheses ().

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